



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations

Summer Semester July 2024

22MCA105 – SOFTWARE ENGINEERING AND TESTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define Software Engineering. What is the difference between software engineering and System engineering.	6	L*1	1	1
	b)	Describe the Software Process Models.	6	L2	1	1
	c)	Mention the Steps followed in Incremental Development	8	L1	1	1
2.	a)	Explain the Problems faced in Waterfall Model.	4	L2	1	1
	b)	List the stages of Testing followed in Software Validation.	12	L1	1	1
	c)	Describe the four fundamental activities that are common to all software processes.	4	L2	1	1
		Unit – II				
3.	a)	Differentiate between Functional and Non-Functional Requirements.	10	L2	2	1
	b)	Outline the Principles followed in Agile Methods.	10	L3	2	1
4.	a)	Explain any two Requirement Elicitation Techniques.	10	L2	2	2
	b)	Describe the Requirement Engineering Process.	10	L2	2	1
		Unit – III				
5.	a)	With a help of use case diagram explain the role of Medical Receptionist.	8	L2	3	2
	b)	Draw a Sequence Diagram for Patient Information System	12	L3	3	1
6.	a)	What are the things you need to consider to develop a system design from concept to object-oriented design.	5	L1	3	1
	b)	Design an interaction between weather station and weather Information system to report weather data using UML.	10	L2	3	2
	c)	With a neat diagram explain System context for weather station.	5	L2	3	1
		Unit – IV				
7.	a)	Mention the six Essentials of Software Testing.	10	L1	4	1
	b)	Outline the Three critical success factors for implementing verification.	5	L2	4	2
	c)	Explain Boundary value analysis in Validation Testing.	5	L2	4	1
8.	a)	Explain error guessing and syntax testing.	10	L2	4	1
	b)	Describe the lesser used methods in Blackbox Testing.	10	L1	4	1
		Unit – V				
9.	a)	Write short note on:				
	i)	Unit testing				
	ii)	Integration testing	10	L2	5	1
	b)	Explain any two testing methods in High level Testing.	10	L2	5	1
10.	a)	Describe Software testing support tools.	10	L2	5	1
	b)	Mention the dangers of having the wrong software testing structure.	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
